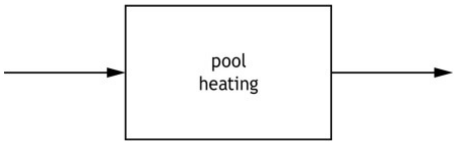
Task 1 — pool heating

The water in the pool complex is heated, to be kept at a constant temperature. The

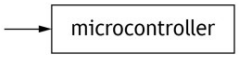
specification for the operation of the pool heating system is given below.

i. The desired temperature of the water is set manually by a pool attendant.
ii. The temperature of the water is constantly monitored. When the temperature falls below the set level, a microcontroller will switch on a heater.
iii. When the temperature of the water rises above the set level, the microcontroller will switch off the heater.

1a (i) Complete the system diagram below to meet the specification. You must clearly show the external input and output. **(2 marks)**



(ii) Complete the sub-system diagram below to meet the specification. You must clearly how the external input and output, all sub-systems, the system boundary and interactions between sub-systems. Any output device requires its own driver. **(6 marks)**


microcontroller